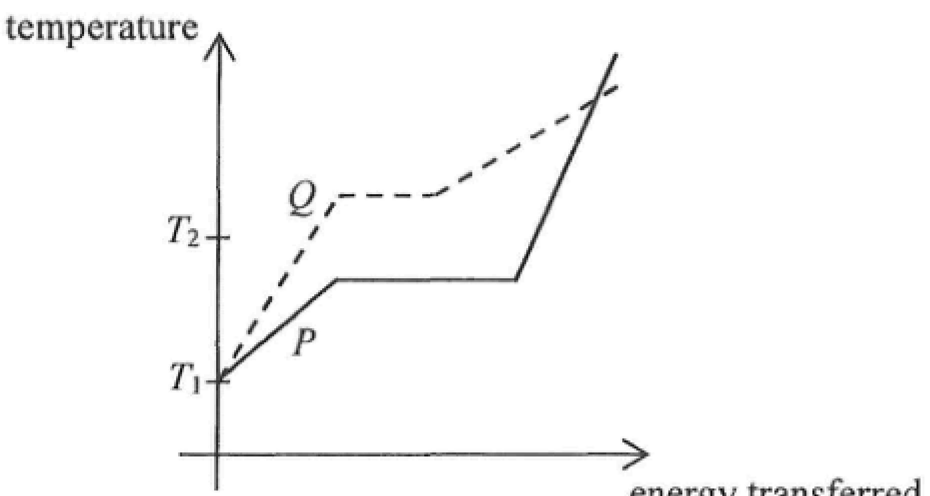
1. Substance P and Q are of the same mass and both are solids at temperature T_1. The graph below shows the variation of temperature of the substances with energy transferred.



Which statement below is correct ?

A. P has a higher melting point than Q.
B. The specific latent heat of fusion of P is smaller than that of Q.
C. The specific heat capacity of solid P is greater than that of solid Q.
D. At temperature T_2, P is gas while Q is solid.

2. If 0.5 kg of water at 50 °C is mixed with 2.0 kg of ice at 0 °C, what is the final temperature of the mixture ?

Given: specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

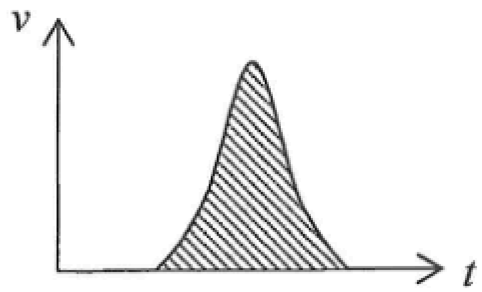
specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}

A. –54 °C
B. 0 °C
C. 10 °C
D. 25 °C

*3. The best vacuum attained in the laboratory has a pressure of about 10^{-8} Pa. Estimate the order of magnitude of the number of air molecules in 1 cm^3 of such 'vacuum' at room temperature.

A. 10^4
B. 10^6
C. 10^8
D. 10^{12}

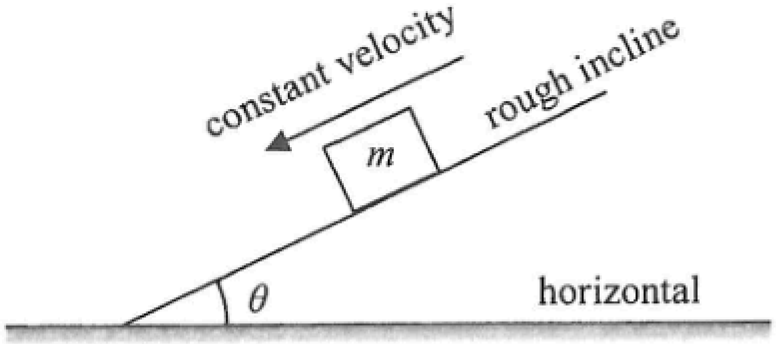
4.



What physical quantity does the shaded area represent in the above velocity-time (v-t) graph ?

A. acceleration
B. change of velocity
C. momentum
D. displacement

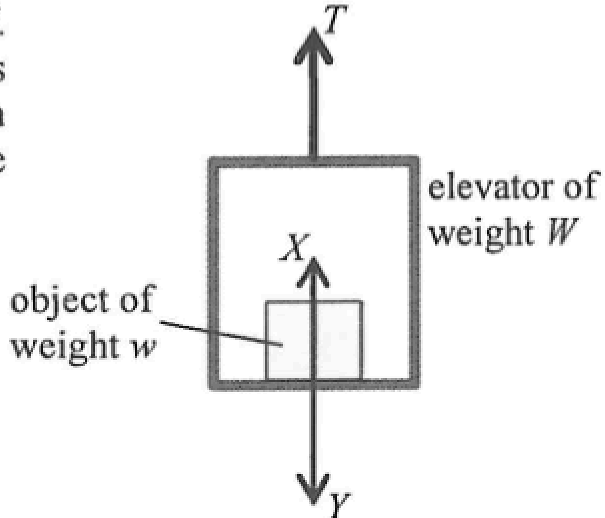
5.



A block of mass m slides down along a fixed rough incline with constant velocity as shown. The incline makes an angle θ with the horizontal. What is the magnitude of the force exerted on the incline by the block ?

A. zero
B. mg
C. $mg \sin\theta$
D. $mg \cos\theta$

6. An object of weight w is placed inside an elevator of weight W. A force T pulls the elevator upward. A **supporting force** X acts on the object while the **floor of the elevator experiences a force** Y due to the presence of the object. Which of the following is an action and reaction pair ?



A. T and $(W + w)$
B. w and X
C. X and Y
D. It cannot be determined as the direction of acceleration of the elevator is unknown.

7. In an experiment, a horizontal force F acts on a block placed on a rough horizontal surface as shown in Figure (a). Figure (b) shows the variation of the block's kinetic energy E_k with the distance s it travelled. Neglect air resistance and assume frictional force unchanged.

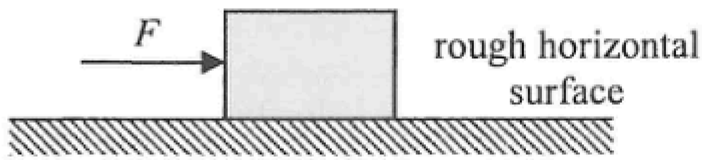


Figure (a)

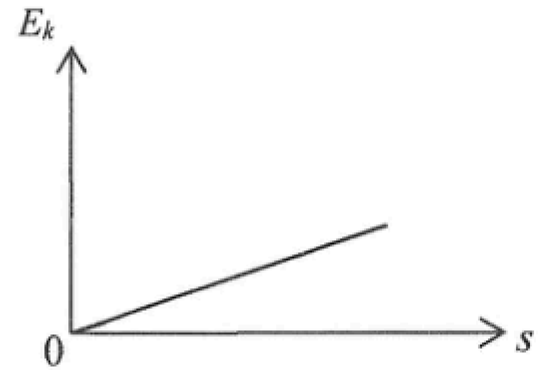
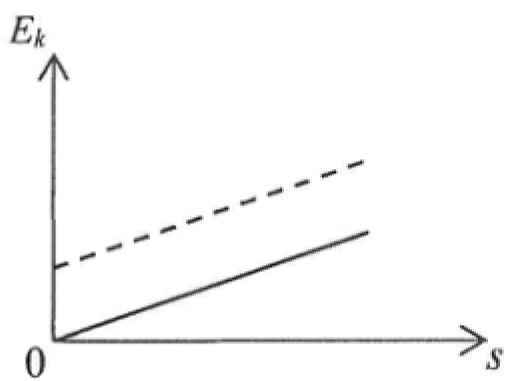


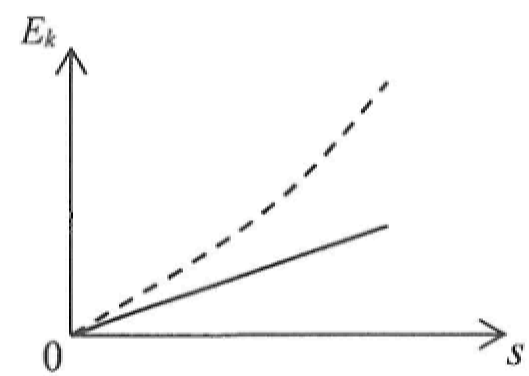
Figure (b)

If the experiment is repeated with force F doubled, which of the following graphs represents the expected result (in dotted line) ?

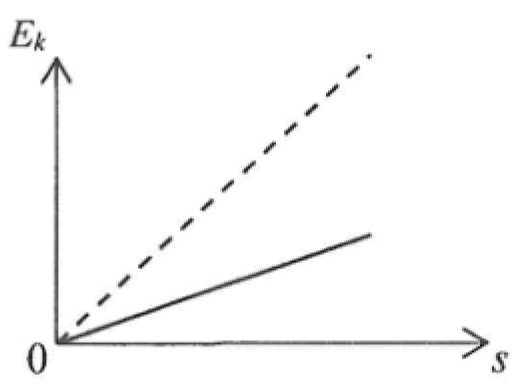
A.



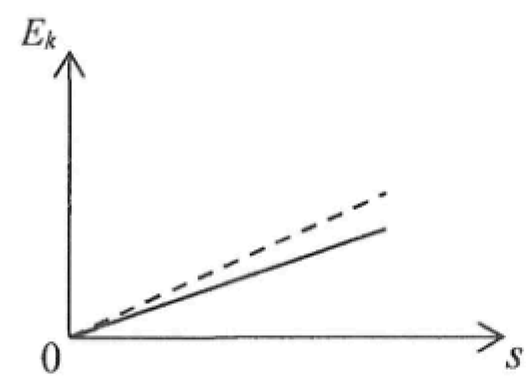
B.



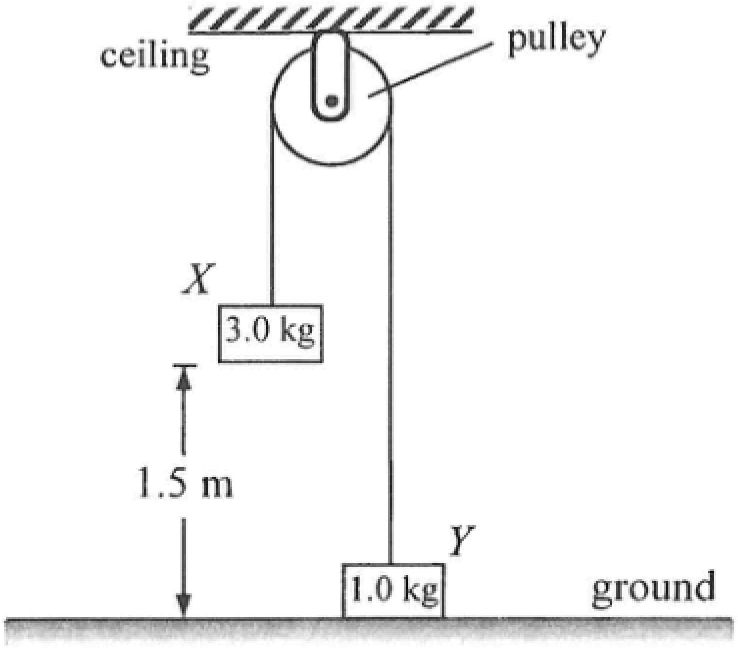
C.



D.



8. Block X and Y of mass 3.0 kg and 1.0 kg respectively are connected by a light inextensible string passing over a smooth light pulley fixed to the ceiling as shown. The blocks are released from rest. Find the time taken for block X to reach the ground. Neglect air resistance.

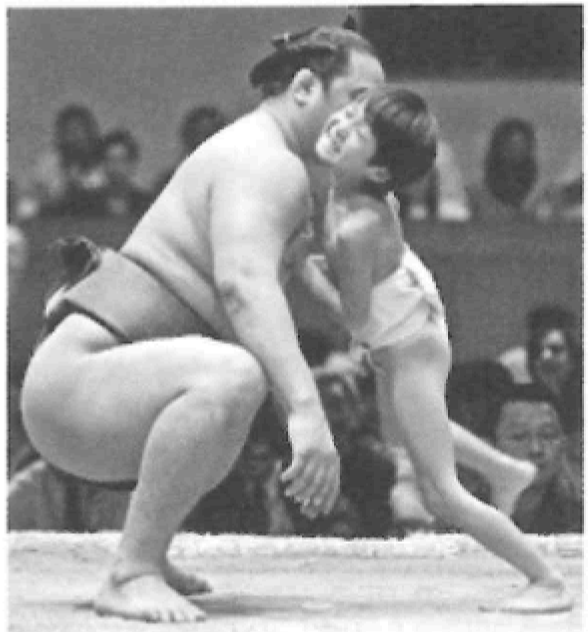


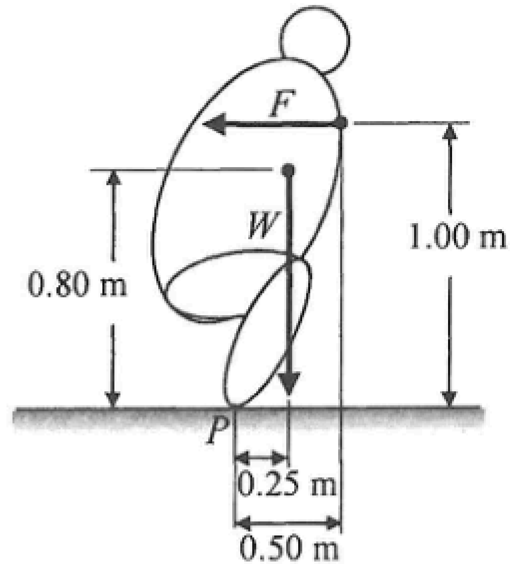
A. 0.55 s
B. 0.64 s
C. 0.68 s
D. 0.78 s

9. A person thought that when he pulls his hair with a large enough force, eventually he will be able to pull himself off the ground. This is a misconception because

A. the pulling force is acting internally within that person but it is not an external force acting on the person.
B. it is impossible to generate a pulling force larger than a person's weight.
C. the pulling force is balanced by the person's weight.
D. it is impossible for the pulling force to overcome the force due to atmospheric pressure.

10. A child tries to push a 150 kg (sumo) wrestler in order to topple him backwards in a game. The simplified diagram indicates the weight W of the wrestler and the horizontal pushing force F acting on the wrestler by the child. P is the wrestler's contact point on the ground.

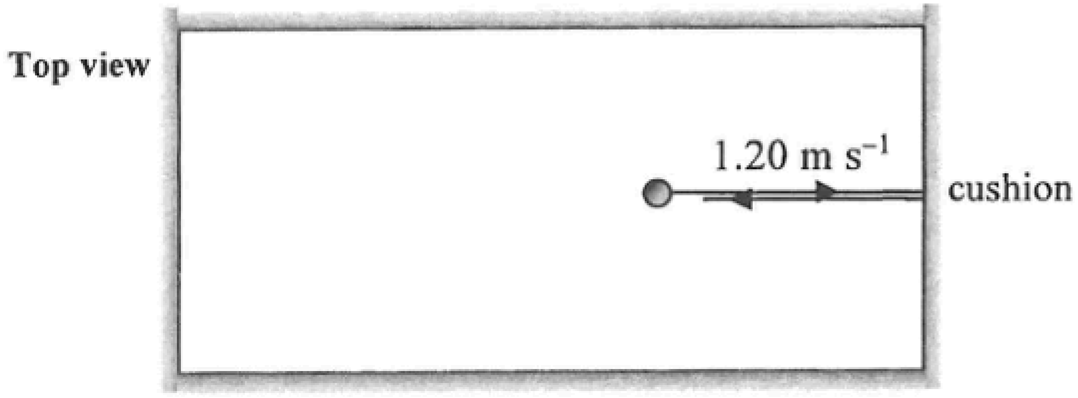




Find the minimum value of F required.

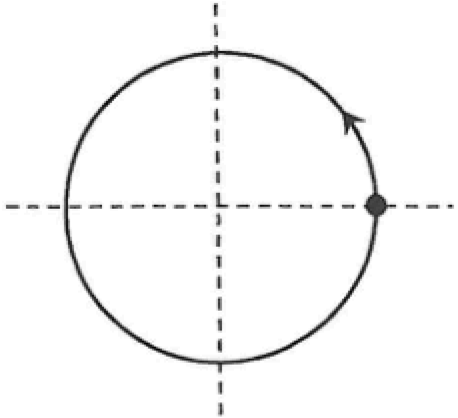
A. 368 N
B. 736 N
C. 1177 N
D. 2354 N

11. On a billiard table, a billiard ball of mass 0.16 kg and moving with a uniform speed of 1.20 m s^{-1} collides elastically with the cushion. It rebounds with the same speed normal to the cushion as shown. If the impact time is 0.0003 s, find the average force acting on the ball by the cushion during collision.

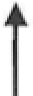


A. 0 N
B. 640 N
C. 1280 N
D. 2560 N

*12.



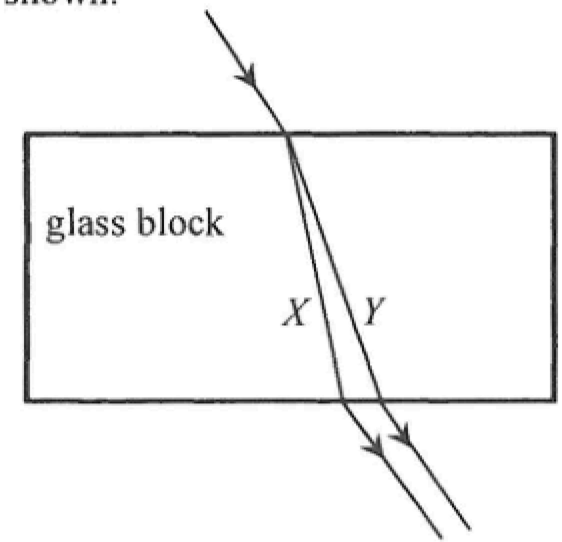
The figure shows a particle performing uniform circular motion. Which of the following correctly indicates the directions of its velocity and acceleration at the instant shown ?

	velocity	acceleration
A.		

*13. A satellite orbits the Earth in a circular orbit of radius r with a period of 24 hours. The period of another satellite which orbits the Earth in a circular orbit of radius $2r$ is

A. less than 24 hours.
B. 24 hours.
C. between 24 hours and 48 hours.
D. more than 48 hours.

14. A light beam consisting of monochromatic light beam X and Y is incident on a rectangular glass block. It splits into two beams as shown.

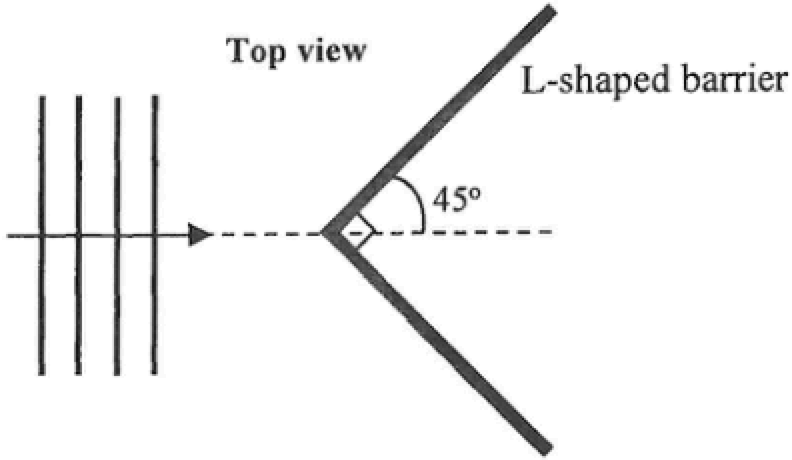


Which of the following comparisons of beam X and Y is/are correct ?

(1) X has a higher frequency.
(2) X has a larger speed in glass.
(3) In the glass block, the critical angle for X is greater.

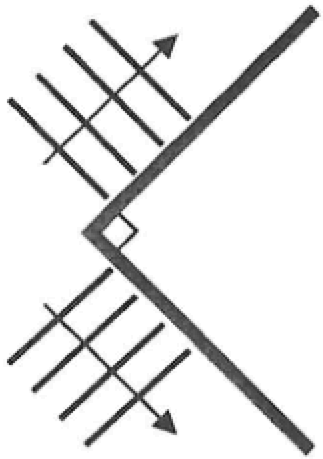
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

15.

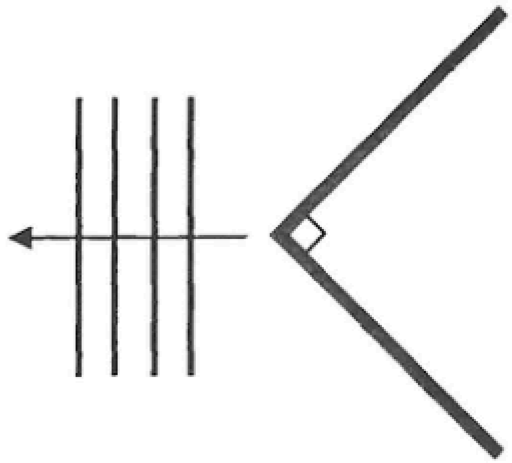


In a ripple tank, a series of plane water waves approach an L-shaped barrier as shown above. Which diagram below best shows the wave pattern reflected from the barrier ?

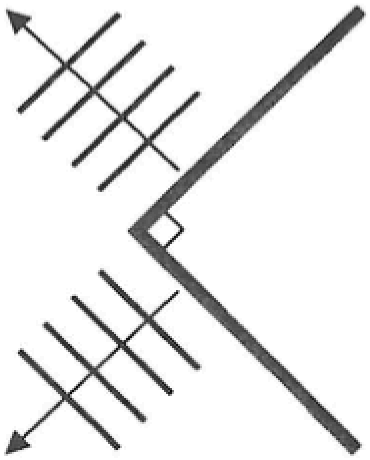
A.



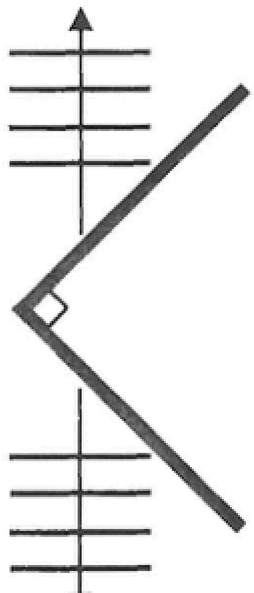
B.



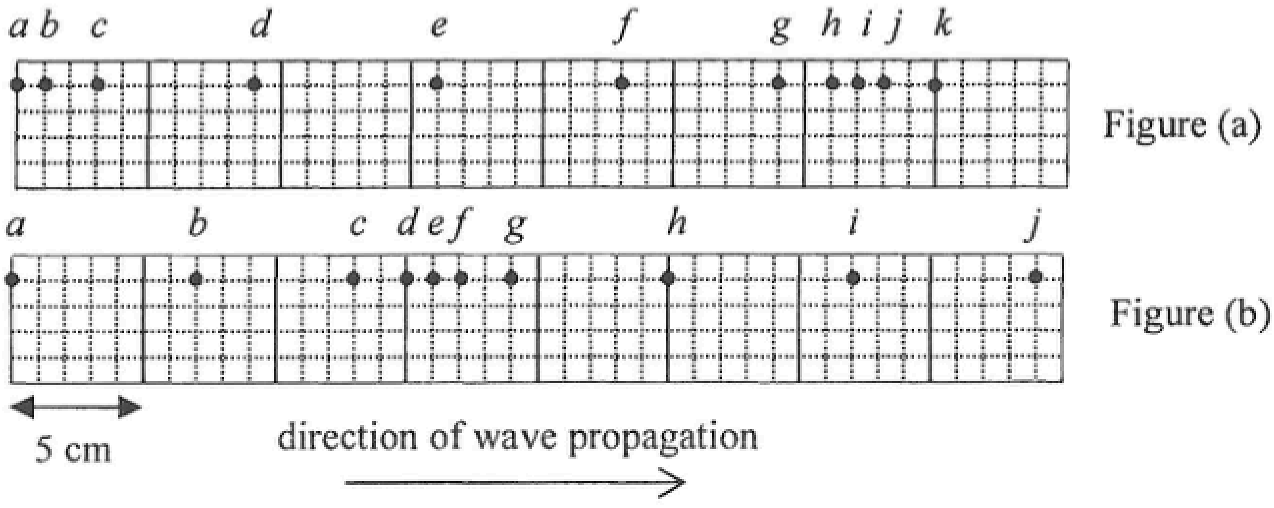
C.



D.



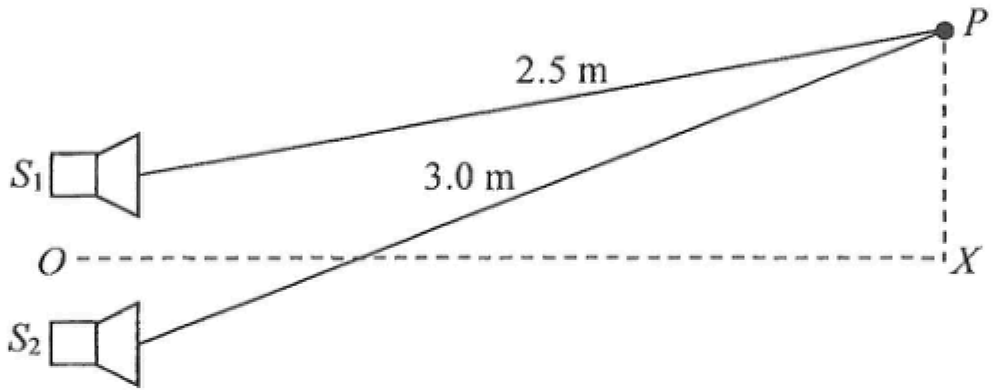
16.



A series of particles is uniformly distributed along a slinky spring initially. Figure (a) and (b) show their positions at two instants when a travelling wave of frequency 10 Hz propagates along the spring from left to right. Which statement below is correct ?

A. Particle *a* is always stationary.
B. Particle *b* and *c* vibrate with different amplitudes.
C. The wavelength of the wave is 16 cm.
D. The smallest time interval for obtaining these two figures is 0.05 s.

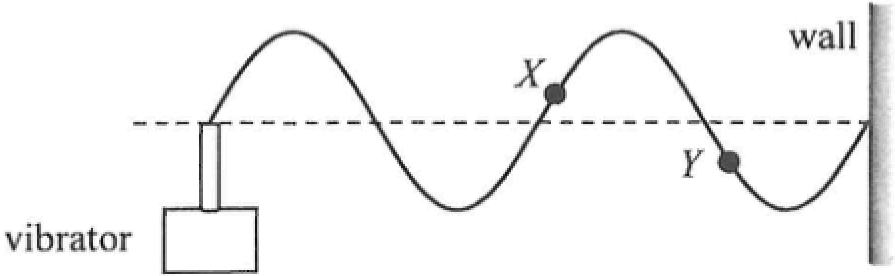
17. Two identical loudspeakers S_1 and S_2 connected to a signal generator produce sounds of wavelength 0.10 m which are in **anti-phase**.



OX is the perpendicular bisector of the line joining S_1 and S_2. What type of interference occurs at X and P respectively ?

	X	P
A.	destructive	constructive
B.	constructive	constructive
C.	destructive	destructive
D.	constructive	destructive

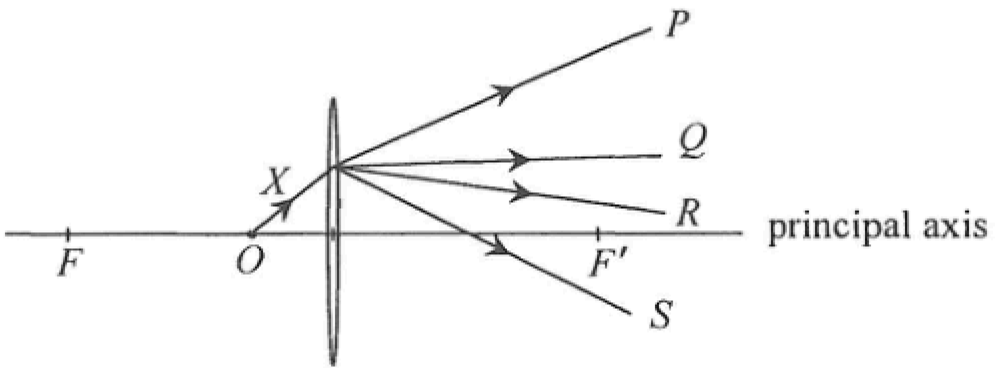
18. A string is tied to a vibrator at one end while the other end is fixed to a wall. The figure shows the stationary wave formed at a certain frequency of the vibrator.



Which statement is correct when the frequency is doubled ?

A. The amplitude of the wave will double.
B. The wavelength will double.
C. The speed of the wave will double.
D. Particle X and Y will become in phase.

19. X is a light ray from an object O placed on the principal axis of a convex lens as shown. F and F' are the foci of the lens. Which light ray best represents the emergent ray ?



A. P
B. Q
C. R
D. S

*20. An object of height 10 cm placed at 20 cm in front of a concave lens forms an image of height 8 cm. Find the focal length of the concave lens.

A. 4.4 cm
B. 8.9 cm
C. 40 cm
D. 80 cm

21. Which description below is **INCORRECT** ?

A. Microwaves are a kind of electromagnetic waves.
B. Microwaves can be observed with the naked eye.
C. Microwaves propagate with the speed of light in a vacuum.
D. Microwaves are used in radar to detect the position of an aeroplane.

22. Figure (a) and (b) are the CRO displays of musical note X and Y using the same time axis and intensity axis.

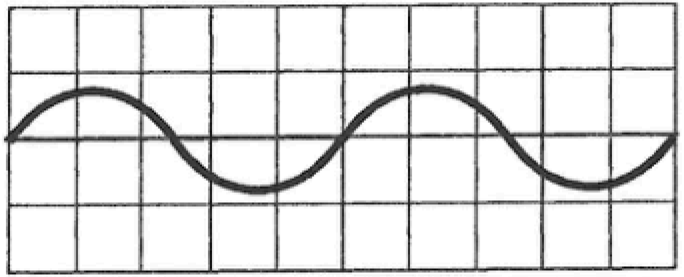
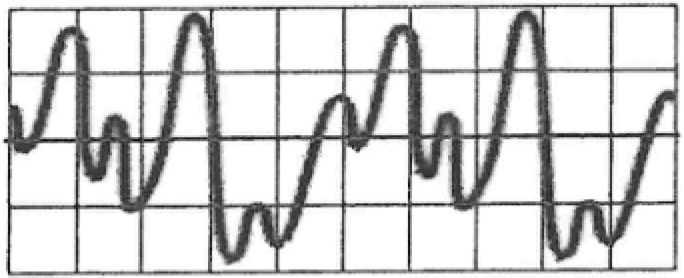
Figure (a)  note X

Figure (b)  note Y

Which comparison of these two notes must be correct ?

	pitch	loudness
A.	the same	the same
B.	the same	not the same
C.	higher for Y	the same
D.	higher for Y	not the same

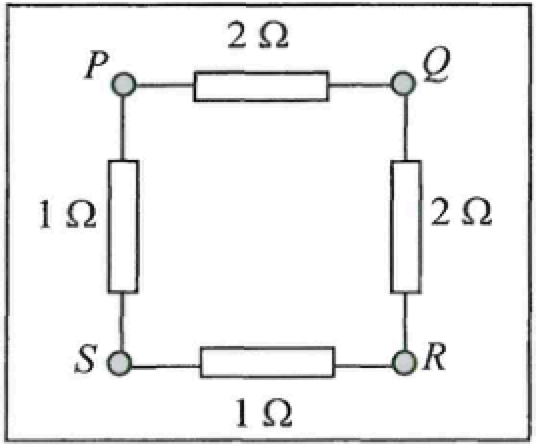
23. Charges $+Q$ and $-2Q$ are fixed at X and Y as shown.

Which description(s) about the electric field along the dotted line is/are correct ?

(1) Between X and Y, the respective electric fields due to the two charges are in the same direction.
(2) To the right of Y, the resultant electric field is always directed to the left.
(3) On the dotted line, there exist two points where the resultant electric field is zero.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

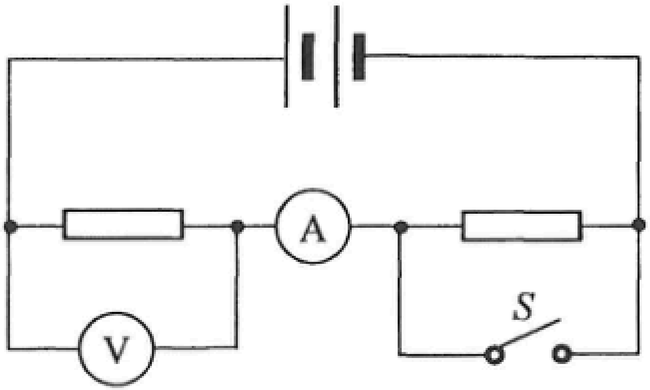
24.



Four resistors are connected to terminals $PQRS$ of a circuit board as shown. Across which pair of terminals below is the equivalent resistance smallest ?

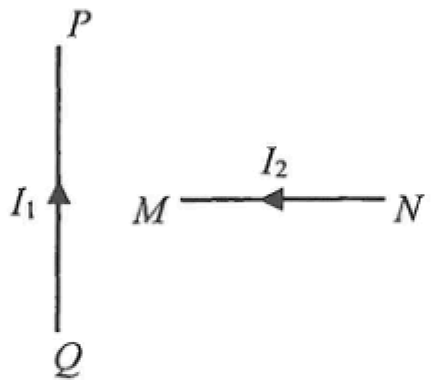
A. PQ
B. PR
C. PS
D. QS

25. In the circuit below, the two resistors are identical. When S is closed, how would the readings of the ammeter and voltmeter change ?



	ammeter reading	**voltmeter reading**
A.	increase	increase
B.	increase	decrease
C.	decrease	increase
D.	decrease	decrease

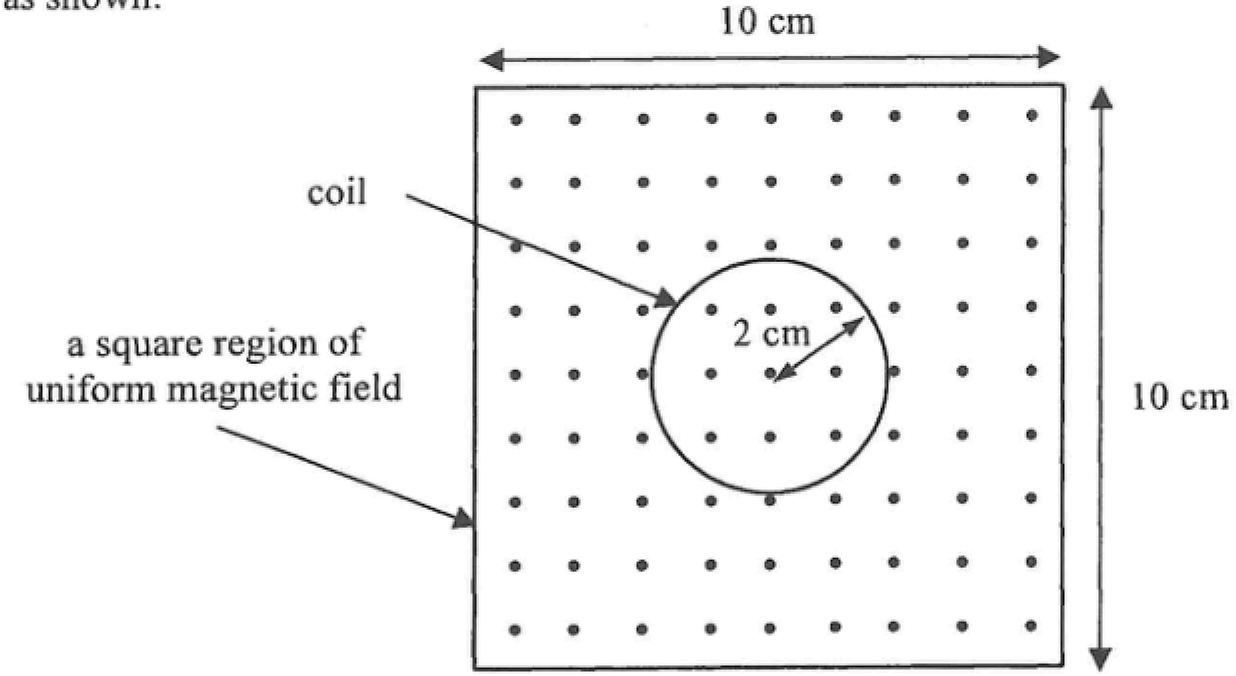
26.



PQ in the figure is a section of a fixed long straight wire carrying a current I_1. MN is a section of another long straight wire carrying a current I_2. MN is perpendicular to PQ and both wires are in the same plane. What is the direction of the magnetic force acting on MN due to PQ ? Neglect the effects of the Earth's magnetic field.

A. pointing out of the paper

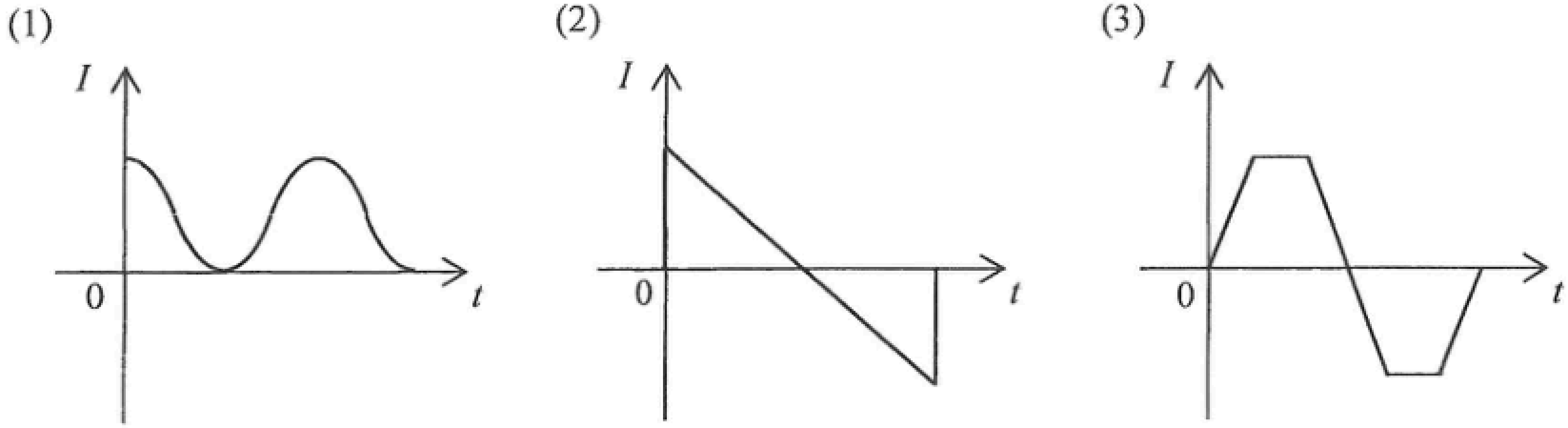
*27. A coil of radius 2 cm is placed inside a square region of uniform magnetic field pointing out of the paper as shown.



The magnetic field decreases uniformly from a strength of 5.0 T to 3.0 T in 0.8 s. What is the magnitude of the e.m.f. induced in the coil ?

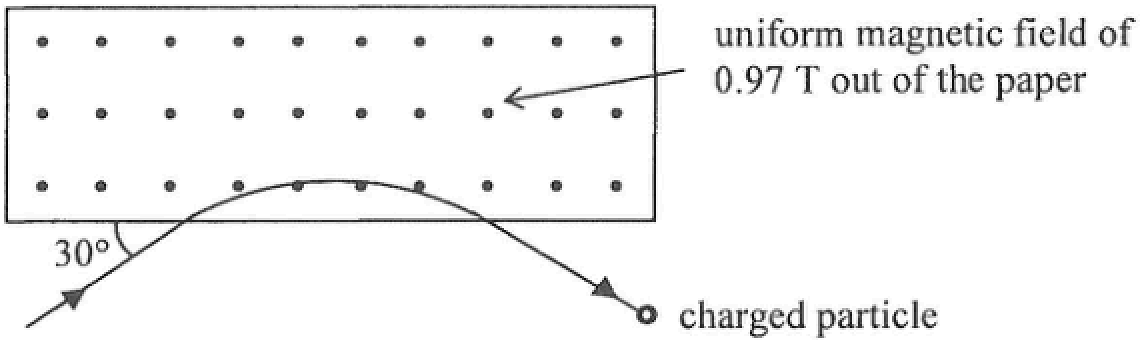
A. 2.5×10^{-2} V
B. 3.1×10^{-2} V
C. 3.1×10^{-3} V
D. 6.2×10^{-3} V

*28. Which of the following current against time (I- t) graphs represent an a.c. ?



A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

*29.



A charged particle enters and leaves a magnetic field of uniform flux density 0.97 T along the path as shown. It experiences a magnetic force of 5.9×10^{-12} N within the magnetic field. Given that the speed of the particle is 1.9×10^{7} m s^{-1}, find its charge.

A. -3.2×10^{-19} C
B. $+3.2 \times 10^{-19}$ C
C. -6.4×10^{-19} C
D. $+6.4 \times 10^{-19}$ C

*30. A power station generates 217 MW of power for transmission to downtown under a voltage of 132 kV via a transmission cable which has an effective resistance of 0.02 Ω km^{-1}. The power loss for a total cable length of 40 km is about

A. 2.16 MW.
B. 6.49 MW.
C. 13.0 MW.
D. 21.6 MW.

31. Which statements about domestic electricity are correct ?

(1) All fuses should be installed in the live wire.
(2) An electrical appliance can work even without an earth wire.
(3) An electrical appliance having a fuse in it can protect people from getting an electric shock.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

*32. It takes 2.0 years for the activity of a sample containing a radioactive nuclide to drop from 800 Bq to 600 Bq. How long will it take for the activity of the sample to drop from 600 Bq to 300 Bq ?

A. 2.0 years
B. 2.5 years
C. 3.0 years
D. 4.8 years

33. In order to locate the cracks in an underground water pipe, a small amount of a radioisotope is added to the water in the pipe. The radioactive substance will leak through the crack to the soil nearby, thus radiation will be detected above ground. Which radioisotope (P, Q, R or S) below is the most suitable for this purpose ?

	isotope	**half-life**	**radiation emitted**
A.	P	66 hours	β
B.	Q	36 hours	γ
C.	R	7.2 hours	α
D.	S	45 minutes	α, β